

Effect of self-assembly on triiodide diffusion in water based polymer gel electrolytes: An application in dye solar cell



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ABSTRACT

The preparation of ordered polymer gels from the amphiphilic block copolymers, Pluronic® F77, P123 and polyethylene glycol in the presence of ionic liquid, iodine and organic additives is presented. At 35%(w/w) concentration these copolymers (F77 and P123) self-assembled into cubic liquid crystalline phase in aqueous solution and characterized by using SAXS and AFM measurements. The effects of micellar aggregation formed by polymers on the ionic transport and triiodide diffusion have been studied by electrochemistry and SANS experiments. The ionic migration or triiodide diffusion through these polymer gels is found to be affected by the PEO/PPO content in the polymer backbone. These gels were successfully employed as an electrolyte in a dye sensitized solar cell. A remarkable solar to electricity conversion efficiency and good stability was obtained using Pluronic® F77 based gel, which is attributed to its thermoreversible sol to gel transition.

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1. Introduction

Polymers have been recently extensively investigated for applications in advanced energy harvesting and storage devices such as dye solar cell, organic solar cell or lithium ion batteries where they play a vital role ranging from active material to neutral backbone [1,2]. In particular, amphiphilic block copolymers are now used as a backbone material in polymer gel electrolytes (PGE) for dye solar cell and lithium ion batteries [3,4] owing to many advantages in terms of safety and easy handling. PGE have unique hybrid structure having cohesive properties of solid and diffusive properties of liquids. PGE also have comparable conductivity as that of liquid electrolytes and have good penetration property. Due to these reasons, PGE are potential substitute for conventional liquid electrolytes. In addition, the use of ionic liquid (IL) and/or water in gel electrolyte will confer a greener and safer approach toward sustainable technology. It has been reported that a thermoreversible gel electrolyte based on the composite mixture of IL and a triblock copolymer (containing 10% polymer in IL) shows better ionic conduction [5].

Photoelectrochemistry for solar to electricity conversion has been extensively investigated for a long time and was accelerated by Gratzel's pioneering work on Dye Sensitized Solar Cell (DSSC) which paved way for the efficient, low cost and cleaner energy from sunlight [6]. Highest 11% efficiency has been reported for DSSC using N719 dye and liquid electrolyte [7]. Traditional organic liquid electrolytes create significant technical problems associated with sealing, corrosion and a lack of stability of the cells, hence awaiting the commercialization of DSSC [8]. Thus, solid and gel based electrolytes such as *p*-type semiconductors (CuI, CuSCN, etc.) [9,10], organic hole conducting materials [11,12], plastic crystal electrolyte [13,14], solid polymer electrolytes and PGE [15,16], etc. have been used as an alternative for liquid electrolytes. The first report on quasi-solid state DSSC appeared in 1995 using poly(acrylonitrile), ethylene carbonate, acetonitrile and NaI with 4.4% power conversion efficiency [15]. After that a series of polymer and polymer composites such as polyethylene glycol, (PEG), poly(ethylene oxide-co-propylene oxide), poly(acrylamide)/PEG, poly(methyl acrylate), poly(vinylidene fluoride-co-hexafluoropropylene) were employed for fabricating solid and quasi-solid DSSC [17–20]. Like solid polymer electrolytes, PGE reduces the efficiency of the DSSC compared to those with liquid electrolytes. This is due to the imperfect contact with the photoelectrode and increase in viscosity upon gelation [21,22]. Because of these drawbacks, most of the

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recent studies have used combination of polymer/copolymer with IL in order to improve the ionic conductivity and the mechanical properties of the PGE [3,4,23–25].

On the other hand all these quasi-solid electrolytes are based on organic solvent and the substitution of these organic solvents in polymer gel with water is one of the important research topics in the context of environment as well as economy. Before 1990, all the DSSC studies were carried out in water based electrolytes with maximum efficiencies 2% and 1.2% at 0.07 sun and 0.5 sun respectively (max. J_{sc} = 0.8 mA/cm²) [23]. Dai et al. reported 1 mA/cm² and 0.45 V using natural dye and water based electrolyte [24]. In 2003, Kaneko et al. reported first quasi-solid DSSC using natural polysaccharides in water containing LiI and I₂ having ~0.6% efficiency [25]. Afterward no efforts have been made to optimize the performance of DSSC using water based PGE. Very recently, Law et al. ruled out the opinion that water is poisonous to DSSC by achieving 2.4% photovoltaic performance of water based DSSC (J_{sc} = 4.7 mA/cm² and V_{oc} = 0.74 V) [26]. To the best of our knowledge, no report has been made on microstructured amphiphilic block copolymer based hydrogel as a redox mediator in dye solar cell as well as the effect of PEO/PPO ratio on triiodide diffusion. Recently we have developed aqueous polymer gels doped with LiI/I₂, which exhibit microcrystalline phase dependent conductivity [27]. The amphiphilic block copolymer, Pluronic® F77 (PEO₅₂-PPO₃₅-PEO₅₂) which self-assembles into liquid crystalline phases such as cubic, hexagonal, lamellar, was used to prepare polymer gel. The result shows that due to interconnected micellar channels in the bi-continuous cubic phase, polymer gel exhibits higher ionic conductivity compare to other crystalline phases [27].

Here we report, the effects of the PEO/PPO ratio and self-assembly in block copolymers on the diffusion of triiodide using cyclic voltammetry at thin layer cell. The variation in triiodide diffusion as a function of PEO/PPO ratio is justified by tracing the location of triiodide species using Small Angle Neutron Scattering (SANS). The diffusion of triiodide and its effect on the performance of DSSC is discussed as a function of mobility of triiodide species.

2. Experimental

2.1. Materials

Pluronic® F77 and P123 were obtained as gift samples from BASF, USA and purchased from Sigma–Aldrich, India respectively and their details are given in Table 1. Polyethylene glycol was purchased from Sigma–Aldrich. D907 dye was received as gift sample from Eversolar Inc., Taiwan (see Supporting Information Fig. S1). Unless and otherwise stated all the chemicals were purchased from Sigma–Aldrich and used as received. All solvents and IL (1-butyl-3-methyl imidazolium iodide, [BMIM][I]) were purchased from Merck, India and used as received. Water used in this study was Milli – Q grade (18 MΩ).

2.2. Methods

2.2.1. Preparation of polymer gels

All the polymer gels were prepared in situ by mixing 500 mM [BMIM][I], 50 mM I₂, 0.5 M 4-tert-butyl pyridine (TBP), appropriate

amount of water and polymer in a glass bottle. These bottles were sealed and stored at lower temperature until homogeneous gels were formed. It is worth to mention that all the gels were composed of 35%(w/w) block copolymer.

2.2.2. DSSC fabrication

Dye coated TiO₂ electrodes were fabricated as per our earlier report [28]. A thin layer of nanoporous TiO₂ was deposited from ethanolic titanium tetra-isopropoxide solution on a cleaned FTO (15 Ω/cm², Solaronix, SA, Swiss) conducting glass with >80% transmittance in visible region, by spin coating at 2000 rpm. This thin layered titania was annealed at 450 °C for 20 min. Titania paste consists of TiO₂ (P-25, Degussa), and ethyl cellulose and α-Terpineol were deposited on above pretreated FTO glass by screen printing technique [29]. The electrodes were fired into a tubular furnace at 500 °C for 30 min and the net thickness of titania film was about 12 μm. Working electrodes were soaked into the TiCl₄ solution for 20 min at 60 °C and sintered in air at 450 °C for 10 min. The electrodes were allowed to cool at 70 °C and immersed into the 30 mM solution of D907 dye (Eversolar, Taiwan) [30], in anhydrous acetonitrile: tert-butyl alcohol (1:1) for 12 h. Electrodes were washed thoroughly with acetonitrile and dried under the stream of nitrogen gas. Counter electrodes were prepared from Platisol solution (Solaronix, SA, Swiss) by spin coating method and rapidly fired into the furnace at 450 °C for 20 min. Gel was dropped onto the dye sensitized electrode and applied two heating and cooling cycles between 10 and 40 °C (in order to get good penetration of gel). By keeping 25 μm spacer, cells were sealed using epoxy adhesive and stored in dark for 12 h prior to measurements. The active cell area was 0.5 cm². For comparison purpose, DSSC was also fabricated with same procedure just by replacing PGE with liquid electrolyte containing 0.5 M IL ([BMIM][I]), 0.05 M I₂, 0.5 M 4-tert-butyl pyridine in acetonitrile.

2.2.3. Characterization of polymer gels and DSSC

Small Angle X-ray Scattering (SAXS) measurements were carried out using the Rigaku SAXS diffractometer equipped with a Gabriel 2D wire detector (sample to detector distance 830 mm and wave length 0.154 nm at IMMM, Le Mans, France). Measurements were collected over a typical time of 10000 s. SANS experiments were carried out on micellar solutions of triblock copolymers, P123 and F77. All the final solutions used for SANS measurements were prepared in D₂O (99.9 atom D%, Sigma–Aldrich, India). This provides very good contrast between micelles and solvent in a SANS experiment. SANS measurements were performed using a fixed geometry SANS instrument with sample-to-detector distance 1.8 m at the DHRUVA reactor, Trombay, INDIA [31]. This spectrometer makes use of a BeO filtered beam which provides a mean wavelength of 5.2 Å and a wavelength resolution of about 15%. The angular distribution of scattered neutrons was recorded using an indigenously built one-dimensional position sensitive detector. The accessible wave transfer ($q = 4\pi \sin 0.5\theta/\lambda$, where λ is the mean wavelength of the incident neutrons and θ is the scattering angle) range of this instrument is 0.015–0.35 Å^{−1}. The solutions were held in 0.5 cm path length UV-grade quartz sample holder with tight fitting Teflon stoppers sealed with parafilm. The liquid fluid mixtures were inserted in 1 mm quartz capillaries, whereas gels were mounted in an aluminum cell having transparent windows. I–V characterizations were carried out using Keithley 2400 source meter and 100 W xenon lamp as light source equipped with band pass filter and light intensity was set to 100 mW/cm² (the light intensity was calibrated using standard Si-photodiode). Electrochemical Impedance Spectra (EIS) of DSSC were obtained by using frequency range from 120 kHz to 50 mHz with 10 mV AC amplitude. Conductivity of the polymer gels were measured using Solartron 1260 (Impedance

Table 1
Description of polymer sample and lattice dimension in polymer gels.

Polymer	Mol.wt. (g mol ^{−1})	% EO	ϕ_{PPO}	a (nm)
P123 (EO ₂₀ -PO ₇₀ -EO ₂₀)	5750	30	0.73	16.6
F77 (EO ₅₂ -PO ₃₅ -EO ₅₂)	6600	70	0.30	25.7
Polyethylene glycol (PEG)	6000	100	–	–

ϕ_{PPO} = Volume fraction of apolar domain.

and Gain Phase Analyzer) by sandwiching the sample between two stainless steel electrodes. Steady state cyclic voltammetry (Solartron 1287 Electrochemical Analyzer) of gels were recorded by sandwiching the gel between two platinum electrodes using 3 M tape as spacer in order to maintain the thickness constant. The voltage was swept from -1 V to 1 V at scan rate 5 mV/s. EIS of the gels were recorded using same thin layer cell. The test cell was swept by AC frequency range from 1 MHz– 0.01 Hz with 10 mV AC amplitude. All the measurements were carried out in duplicate and their mean values are reported here. ^1H NMR and IR measurements were carried out using Bruker Avance 400 (Switzerland) and ABB Inc. (Canada) respectively. Atomic Force Microscope (AFM) (Nanosurf) with resolution 1 nm was used to carry out measurement. All scans were taken at room temperature in air. The glass substrate was carefully cleaned by using ultrasonic treatment with base, acid aqueous solution and alcohol, and then dried in an oven at 100°C . The thin film of F77 based gel was prepared by a dip-coating method (withdrawing speed: 2 mm S^{-1}) on glass substrate and the surface topography and roughness of the thin film was characterized and analyzed by AFM. Prior to AFM measurements, the gel coated films were dried at 100°C for 10 min. Roughness and fractal measurements on the sample were carried out using built-in software on the AFM. Both area measurements and line profiles were used to calculate surface roughness and diameter. In all the AFM scans, there are 500×500 data points across the areas of the scan, and contact scans were made.

3. Results and discussion

3.1. Self-assembly and ionic migration in polymer gels

In water, the amphiphilic block copolymers, F77 and P123 self-assemble into bi-continuous cubic, 2D-hexagonal and lamellar liquid crystalline phases [27,32]. In presence of a salt (LiI/I_2), liquid crystalline phases of F77 shows following order of conductivity in water; cubic > lamellar > 2D hexagonal [27]. The highest conductivity in cubic liquid crystalline phase has been attributed to the presence of interconnected bi-continuous channels in the cubic phase. On the basis of this, here we have selected the 35%(w/w) concentration of block copolymer which corresponds to the cubic phase for both copolymers.

Fig. 1(a) shows the SAXS profile of 35%(w/w) F77 and P123 amphiphilic block copolymers in presence of $[\text{BMIM}][\text{I}]/\text{I}_2/\text{water}$ at 30°C . The sequence of Bragg reflections defined by their Miller indices (hkl) is used to identify the nature of a particular phase. In the SAXS pattern (Fig. 1(a)), we have observed the Bragg reflections, follow sequence, $\sqrt{2}:\sqrt{4}:\sqrt{6}:\dots$ and $\sqrt{3}:\sqrt{2}:\sqrt{8}:\sqrt{11}:\dots$ in q space for F77 and P123 respectively. These Bragg reflections can be indexed as (110), (200), (211) ... and (111), (200), (220), (222) and the sequences are typical of bi-continuous $Im3m$ and $Fm3m$ cubic crystalline phases respectively with spherical micelles in shape [32,33]. However, the binary phase diagrams of F77 and P123 in water (in absence of IL/salt) nevertheless exhibit normal $Im3m$ and $Fm3m$ cubic phases. The presence of IL and salt induce the transformation from a cubic to a bi-continuous lyotropic liquid crystalline phase.

The refinement of the peak positions for both structures allow us to extract the lattice parameter, a of these gels and the obtained values are reported in the last column of Table 1. The value of a for P123 is lower than for the F77 gel. One of the main reasons for this decrement comes from the fact that the spherical micelles of P123 are closely packed. This is attributed to the lower degree of hydration in P123 micelles due to the high value of apolar volume fraction relative to F77 block copolymer (Table 1). Moreover, in order to see the structural organization of self-assembled block

copolymer in presence of salt/IL, a thin film (prepared using dip coating) of F77 gel was prepared on glass substrate and characterized by AFM. Accordingly, Fig. 1(b) is the AFM image of casted gel which shows the presence of domain with short range ordering in the film and the calculated roughness is about 2 ± 0.1 nm. The black dark spots in the image prove the existence of a circular area ranging from 3 to 5 nm. This might be the core of F77 micelles consisting of hydrophobic PPO moieties. The contrast of this area to rest of the film (the shell composed of PEO) could be due to the penetration of relatively hydrophobic I_3^- into the PPO core. The obtained diameter of these spots is $\sim 3 \pm 0.5$ nm which matches well with the core radius of F77 micelles obtained by SANS measurements (details can be found in successive discussion on SANS results). The vertical lines in the image may be due to the line flattening smoothing process. The inset of Fig. 1(b) shows the FFT (Fourier Transformation and Filtering) of the AFM pattern. It consists of four spots surrounding the central one at equal distance thus confirming the cubic crystalline nature of the F77 gel.

The ionic conductivity of the polymer gels has been probed using AC impedance method by sandwiching them between two stainless steel disk electrodes. Conductivity data are collected at different temperatures and fitted into the Arrhenius type equation. Plot of $\ln \sigma$ Vs $1/T$ has been shown in Fig. 2, which has linear behavior of conductivity with temperature. The extracted activation energies (E_a) and conductivity at 30°C are depicted in Table 2. From these parameters, it is clear that the gelation of polymer has little influence on the ionic conductance of PGE and the conductivity trend follows $\text{P123} > \text{F77} > \text{polyethylene glycol (PEG)}$. This anomalous behavior is due to the variation in PEO/PPO ratio amongst the block copolymers and their self-assembly in aqueous solution of salt/IL. When F77 and P123 block copolymers were self-assembled in higher order structures, ethylene oxide and propylene oxides form hydrophilic and hydrophobic domain respectively. Due to less number of PEO blocks in P123 gel, the degree of hydration per PEO chain is high compared to F77. This facilitates the movement of ions in aqueous media.

In order to support this mechanism and to check the feasibility for the use of these polymer gels in dye solar cells, it is necessary to study the diffusion of triiodide species (the mixture of IL ($[\text{BMIM}][\text{I}]$) and iodine generates triiodide species). The diffusion coefficients of triiodide have been measured using cyclic voltammetry at thin layer cell. The apparent diffusion coefficients (D^*) of the ions can be expressed in terms of effective diffusion coefficient using Dahms-Ruff equation [34],

$$D^* = D + k_{ex}\delta^2 C_0/6 \quad (1)$$

where the first term i.e. D denotes the effective diffusion coefficient and second term, $(k_{ex}\delta^2 C_0/6)$ denotes electron hopping self-exchange mechanism. In liquid and gel electrolytes, electron self-exchange is quite fast. Hence the contribution of the second term is minor, and therefore neglected. In such conditions, apparent diffusion coefficient is directly correlated to the effective one. Fig. 3 shows the steady state cyclic voltammogram of different polymer gels and the apparent diffusion coefficient of triiodide species was calculated using following equation [35,36],

$$D_{\text{I}_3^-} = \frac{J_{\text{lim}} L}{2nFC_{\text{I}_3^-}} \quad (2)$$

where $D_{\text{I}_3^-}$ is the diffusion coefficient of triiodide ion, J_{lim} is the limiting current densities, L is the thickness of cell, n is number of electron transferred ($n = 2$), F is faraday constant (96480 Coulomb), $C_{\text{I}_3^-}$ is the concentration of triiodide in gel.

Last column of Table 2 reveals that the triiodide ions can diffuse faster in case of microcrystalline ordered P123 and F77 gels compare to order less polyethylene glycol gel explains the same trend

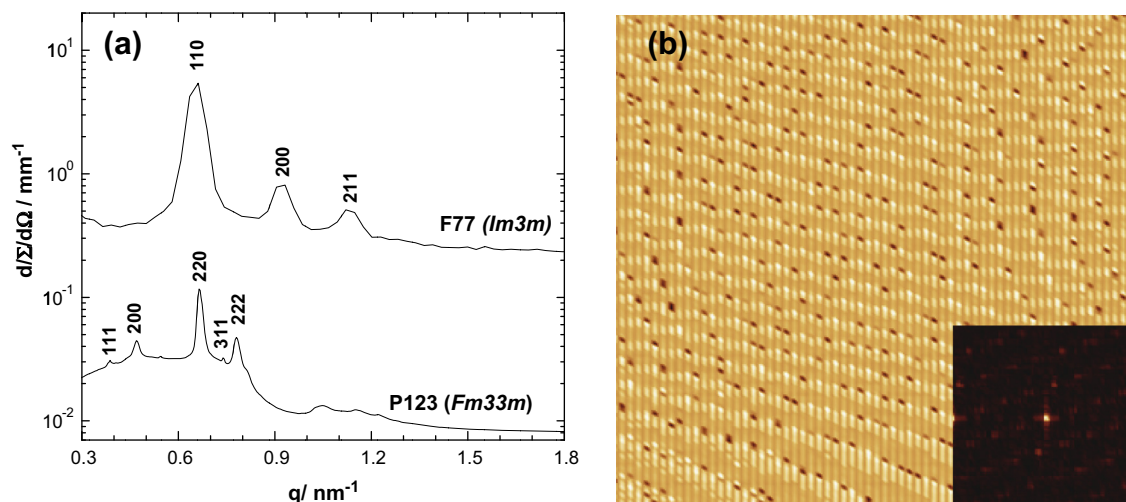


Fig. 1. (a) SAXS pattern of P123 and F77 block copolymer gels in aqueous solution in presence of IL/I₂ (pattern of F77 gel is shifted by a factor of 10), (b) AFM image (500 nm × 500 nm) of F77 gel coated onto glass substrate and inset shows the FFT image of selected area.

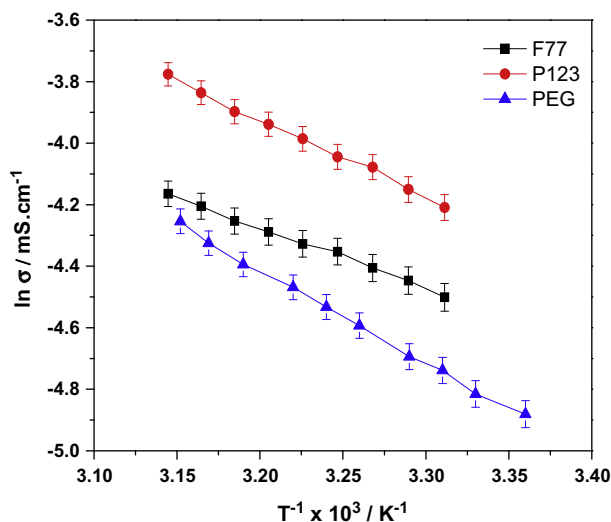


Fig. 2. Arrhenius plots of the various polymer gels containing salts.

Table 2

Conductivity, σ , activation energy, E_a and diffusion coefficient, D_{I_3} of triiodide for various polymer gels.

	σ (mS cm ⁻¹)	E_a (kcal mol ⁻¹)	D_{I_3} (cm ² s ⁻¹)
P123 based gel	14.8	5.0	4.3×10^{-7}
F77 based gel	11.1	3.8	3.4×10^{-7}
PEG based gel	8.8	6.0	2.6×10^{-7}

like in conductivity. These estimated values of diffusion coefficients for all the polymer gels are comparable with those reported ($\sim 1 \times 10^{-7}$ cm² s⁻¹) by other groups [3,17] for pure ionic liquid and polymer gel based electrolytes.

In order to understand the mechanism of I_3^- diffusion and conductivity, it is essential to know the exact location of I_3^- along with other species inside the micellar framework. In view of this, SANS measurements of 15%(w/w) P123 and F77 in aqueous solutions were made at 30 °C. Fig. 4 represents the SANS curves of F77 and P123 aqueous solution in presence of I^-/I_2 .

Strong correlation peaks are observed in both cases and they found at higher q values for F77. This indicates that the micelles

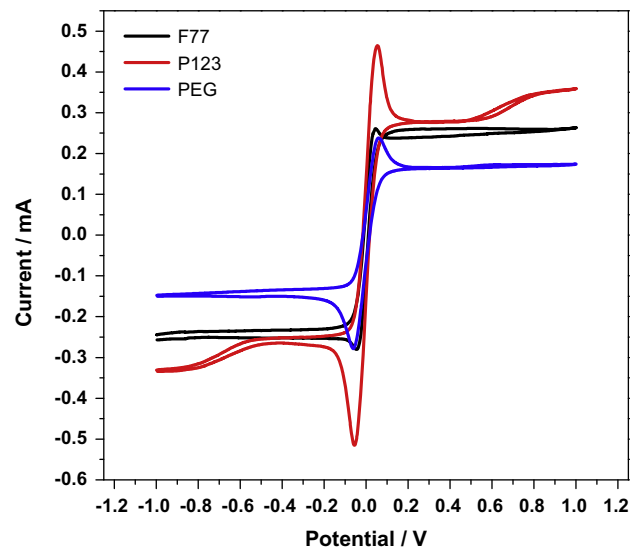


Fig. 3. Steady state cyclic voltammogram of polymer gels at thin layer symmetrical cell. Scan rate is 5 mV/s, $T = 30$ °C.

are smaller in size and have higher intermicellar interaction compared to micelles of P123 in D₂O. This is supported by the parameters obtained from the fitting of the SANS profiles, based on spherical core-shell model (Table 3). From Table 3 it is revealed that P123 micelles possess larger core radius as compared to F77 and thus possess lower volume fraction. Moreover, for PEO-PPO type of triblock copolymers, it is well known that the higher hydration of PEO induces the shrinkage in PPO chains and thus, reduces the core radius.

However, upon addition of (500 mM $I^-/50$ mM I_2) salt in aqueous solution of F77 block copolymer, no significant change in micellar parameters is observed, i.e. R_c , R_{HS} and R_s almost remain unaltered. On the contrary, drastic changes in micellar dimensions for P123 were observed in presence of I^-/I_2 . A decrease in R_c and N_{agg} suggests that the addition of salts induces de-micellization of the P123 micelle. This might be due to the presence of I_3^- species inside the core as well as at the PEO/PPO interface, which reduces the micellar core radius along with R_{HS} , and thus volume fraction. This is more justified as the mixture of I^- and I_2 produces I_3^- which have

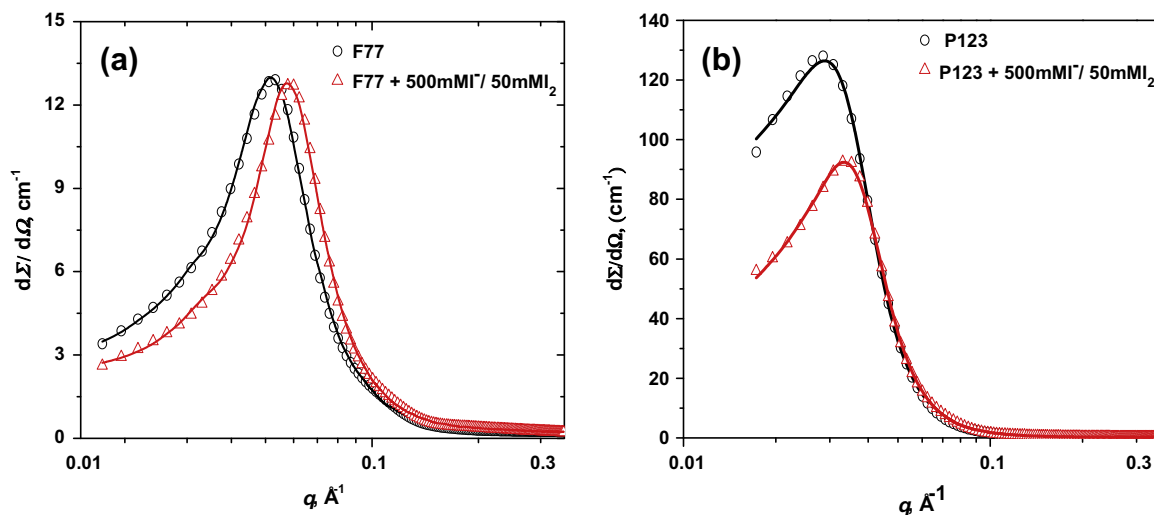


Fig. 4. SANS profile for 15%(w/w) of (a) F77 and (b) P123 in D₂O with and without I[−]/I₂.

Table 3

Parameters obtained from SANS analysis, scattering length density of medium, ρ_M , core radius, R_c , scattering length density of core, ρ_{core} , radius of shell, R_s , polydispersity, volume fraction, ϕ , aggregation number, N_{agg} .

System	$\rho_M \cdot 10^{10}$ (cm ^{−2})	R_c (Å)	$\rho_{core} \cdot 10^{10}$ (cm ^{−2})	R_s (Å)	$\rho_{shell} \cdot 10^{10}$ (cm ^{−2})	R_{HS} (Å)	Poly disp.	Volume fraction, ϕ	N_{agg}
15%(w/w) F77 + D ₂ O	6.38	29.5	0.344	19.7	6.37	53.6	0.299	0.243	38
15%(w/w) F77 + 50 mM I ₂ /500 mM I [−] + D ₂ O	5.80	29.1	0.335	18.4	6.36	49.5	0.206	0.268	36
15%(w/w) P123 + D ₂ O	6.38	49.2	0.344	26.2	6.36	78.4	0.268	0.190	135
15%(w/w) P123 + 50 mM I ₂ /500 mM I [−] + D ₂ O	5.80	44.4	0.312	20.0	6.26	70.5	0.260	0.223	99

relatively less solubility in aqueous media. These moieties prefer to reside in apolar domain (in our case PPO) or PPO/D₂O interface compared to highly hydrated region (PEO domain) [26,37].

We were unable to fit the scattering curves in which the scattering length density of the PPO core is fixed to the one found in pure water ($\sim 0.344 \times 10^{10}$ cm^{−2}). A clear decrease in this parameter is observed upon on addition of I[−]/I₂, i.e. either I₃[−] (I[−] + I₂ ↔ I₃[−]) diffuses inside the core or the PPO density per block decreases in a PPO melt. The presence of I₃[−] in PPO core is more probable since I₃[−] has relatively less solubility in aqueous media. By making hypothesis that PPO blocks and I₃[−] molecules do not change in mean volume when mixing, we can find a fraction of I₃[−] in the core from the fitted scattering length density values, ρ_c using following equation,

$$\rho_c = \phi_{I_3} \rho_{I_3} + (1 - \phi_{I_3}) \rho_{PPO} \quad (3)$$

where ϕ_{I_3} , ρ_{I_3} and ρ_{PPO} are the volume fraction of triiodide in the core, scattering length density of triiodide (0.270×10^{10} cm^{−2}) and PPO (0.344×10^{10} cm^{−2}) respectively. The ϕ_{I_3} values obtained from above equation are 0.131 and 0.467 for F77 and P123 polymer gels respectively, which corresponds to 1.3% and 46.7% of I₃[−] (from 50 mM) is present inside the core. This indicates that PPO core has affinity for triiodide and because of channel like interconnected network, the mobility of ions increases. In case of F77 gel, the mobility of triiodide was hindered due to the smaller size of micellar core and as a result conductivity of F77 gel is lowered. Thus, highest conductivity in P123 gels can be due to the larger size of PPO domain, through which the diffusion of I₃[−] from interconnected micellar channel is easy as compared to F77 gel. In case of hydrophilic PEG gel, due to heavy hydration and lack of ordered structure, the ion migration through gel has been retarded which results into lowering of ionic conductivity. Therefore, finally we can say that the presence of self-assembled structure allows free movement of ions. This fact supports our conclusion that the self-assembly of block

copolymers plays a vital role in the migration of redox mediator. This mechanism has been further supported by Infrared (IR) and ¹H NMR spectroscopy (for details please see [supplementary information](#)).

3.2. Application of polymer gels as electrolytes in DSSC

Photovoltaic characterizations of the DSSC fabricated using the polymer gel as an electrolyte has been carried out. Fig. 5 shows I–V curves obtained for the DSSC assembled with different PGEs as well as liquid electrolyte and parameters calculated from these are shown in Table 4. Devices A, B, C and D consist of liquid, F77,

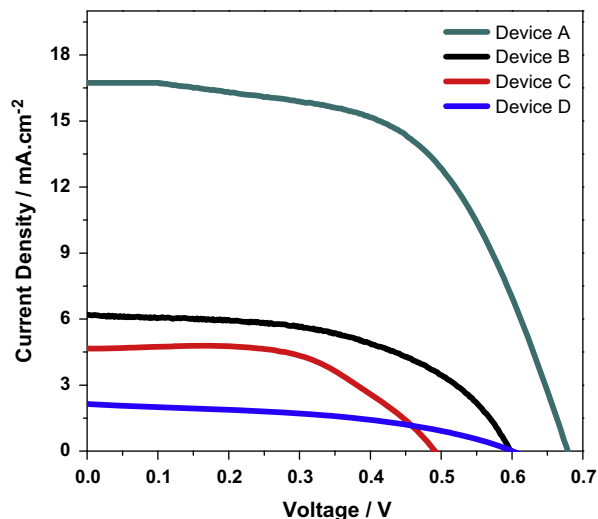


Fig. 5. I–V characteristics of DSSC based on different gels as an electrolyte.

Table 4
Photovoltaic performance parameters of dye solar cells.

Device	Electrolyte	J_{sc} (mA/cm ²)	V_{oc} (V)	FF (%)	η (%)	R_{ct} (Ohm) ^e	τ_{ele} (ms) ^e
A	LE ^a	16.8	0.676	54	6.5	2.75	41
B	PGE 1 ^b (F77)	6.2	0.595	53	2.1	4.12	50
C	PGE 2 ^c (P123)	4.6	0.491	57	1.3	5.62	132
D	PGE 3 ^d (PEG)	2.1	0.600	44	0.6	–	–

^a 2 M IL, 50 mM I₂, 0.5 M TBP in acetonitrile.

^b 2 M IL, 50 mM I₂, 0.5 M TBP and water in F77.

^c 2 M IL, 50 mM I₂, 0.5 M TBP and water in P123.

^d 2 M IL, 50 mM I₂, 0.5 M TBP and water in PEG.

^e Parameters obtained from impedance measurements.

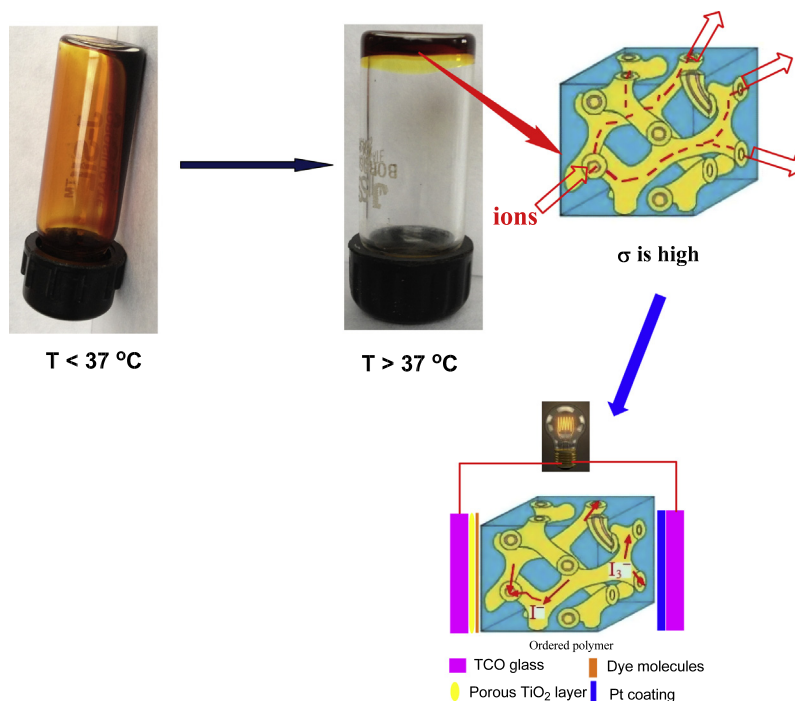
P123 and PEG based electrolytes respectively. These data set forth a clear enhancement in the photocurrent when ordered amphiphilic block copolymer based gels are used instead of non-ordered PEG based electrolytes. In block copolymer based gels, J_{sc} increases from 4.6 to 6.2 mA/cm² and overall efficiency reaches to 2.1% (which is quite comparable in the category of pure water based PGE) when we switch over from P123 to F77 gels. The obtained value is quite comparable with those recently reported by Law et al. using water based liquid electrolytes (~2.4%) [26]. This performance is unexpected considering the enhanced ionic conductivity and higher triiodide diffusion in case of P123 gel as discussed before.

Nonetheless this anomalous behavior can be explained by considering the physical state of the so formed gel electrolytes. P123 forms a quasi-solid gel having very high viscosity without any flow properties while F77 gel shows remarkable structural polymorphism i.e. a thermoreversible sol to gel transition. By lowering the temperature of the F77 gel, a conversion into the simple micellar solution having reduced viscosity occurs [27]. So at lower temperature F77 gels transform into a solution which can thereby diffuse into the nanopores of TiO₂. The solution can transform again into a solid phase upon increase in temperature, which is not the case for P123 gels. The schematic representation of thermoreversible PGE based on F77 block copolymer and its use in DSSC is shown in Scheme 1.

Poor contact of the electrolyte with photoelectrode would lead to diminished current density. Moreover, higher PPO content in the P123 gel would hinder the accessibility of 4-tert-butyl pyridine which is responsible for V_{oc} enhancement, results in the overall reduction in performance of DSSC. Stability test for the efficient DSSC was carried out over 500 h time period (Fig. S4 Supporting Information) and it shows that there is only ~3% reduction in the overall performance of DSSC based on F77 PGE. On the contrary, dye desorption has been observed in case of polyethylene glycol gel which may be due to highly hydrated PEG chains, while it was not the case for P123 and F77 block copolymer based PGE.

Electrochemical Impedance technique probes three main electrochemical processes occurring during the operation of DSSC and its response is plotted in the form of Nyquist plot (Fig. 6). The typical Nyquist plot of DSSC generally represents three semi-circles corresponding to different electron transfer processes occurring inside the DSSC. The higher frequency arc reflects the reduction of triiodide at Pt loaded FTO surface, the middle arc reveals electron transfer at electrolyte/Dye/TiO₂ interface, while the lower frequency arc reflects mass transport through the electrolyte [38,39]. Fig. 6 confirms the basics functions; injection, regeneration and transport of electron in DSSC where all water based PGEs are used.

Impedance data have been fitted into equivalent model circuit (see supporting information, Fig. S5). Charge transfer resistance



Scheme 1. Schematic representation of F77 based PGE and its use in DSSC.

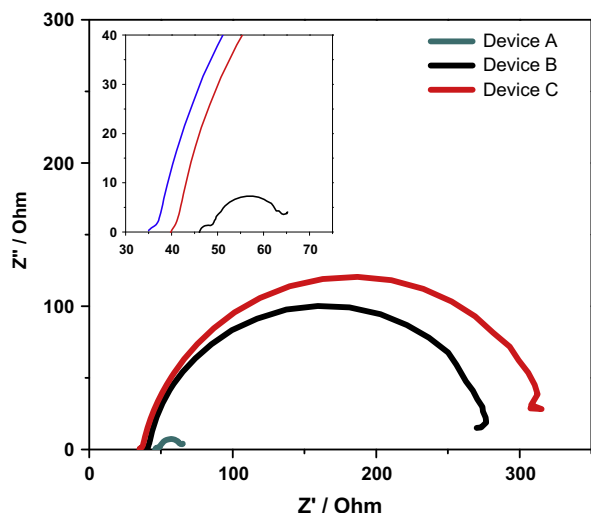


Fig. 6. Nyquist plots of different DSSCs.

(R_{ct}), an important parameter which reveals the electron transfer resistance at Pt/electrolyte interface, was approximated by the diameter of high frequency semicircle. Values of R_{ct} for different DSSC are given in second last column of Table 4.

From the table it is clear that charge transfer resistance in the devices A and B is less as compared to C. This is due to the fact that electrolyte with lower viscosity makes better contact with the platinum electrode thereby decreasing the resistance of electron transfer. Another parameter, electron lifetime, τ_{ele} , has been calculated from the peak frequency (ω_{max}) of second semicircle. Electron life time gives an idea about the time which an electron spends in the titania matrix before reaching to the collector or recombining with the triiodide. Values of τ_{ele} are depicted in Table 4, which indicates that the electron lifetime increases as the viscosity of gel increases. Increasing viscosity restricts the flow properties hence gel cannot be penetrated into the titania film which in turn retards the recombination. Due to the reduction in the recombination processes electron gets sufficient time in the titania film leading to increase in τ_{ele} . But this was not the case with F77 gel since it has thermoreversible nature, and penetration of it into the TiO_2 matrix becomes easy and thereby increasing the electron recombination and lowering its lifetime.

4. Conclusion

In summary, we have shown that we could prepare highly ordered polymer gels from P123 and F77 amphiphilic block copolymers using IL, [BMIM][I] and iodine. The high conductivity of these PGE is attributed to the easy diffusion of I_3^- ions through the bi-continuous channel like structure of these polymers. Triiodide diffusion through such polymer gels depends on the ratio of hydrophilic and hydrophobic domains. These ordered water based PGE are used as a redox mediator in dye sensitized solar cell and a remarkable 2.1% solar to electricity conversion efficiency was achieved using D907 dye as a sensitizer. Among all PGE studied here, F77 gels are promising candidate as an electrolyte due to their thermo reversible nature. The efficiency of the DSSC device can be tuned by selecting appropriate solvent and polymer. The ordered water based PGE can be a good substituent for an organic solvent based polymer gels for advance electrochemical devices such as lithium ion batteries, fuel cells, solar cell.

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Appendix A. Supplementary material

Electronic Supplementary Information (ESI) available: The specifications of dye molecule, IR study, ^1H NMR study, stability graph, circuit model for impedance fitting and Impedance study at thin layer cell. The detailed analysis of Small Angle Neutron Scattering measurements is also given. Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jcis.2014.03.047>.

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